MIL-G-20078C
7 July 1983
SUPERSEDING
MIL-G-20078B
10 June 1963
(See 6.5)

MILITARY SPECIFICATION

GASKETS FOR BALLISTIC DOORS AND HATCHES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metal inserted rubber gaskets and two-line rubber gaskets for ballistic doors and hatches (see 6.1).

1.2 Classification. The gaskets shall be of the following types, as specified (see 6.2.1):

Type A-1 - Metal inserted; straight section.
Type A-2 - Metal inserted; corner section.
Type B - Two-line rubber.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-289 - Visual Inspection Guide for Rubber Sheet Material.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 429 - Rubber Property - Adhesion to Rigid Substrates. (DoD adopted)
D 1081 - Evaluating Rubber Property - Sealing Pressure. (DoD adopted)
D 2240 - Rubber Property - Durometer Hardness. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Material. The material shall be as specified hereinafter.

3.2.1 Rubber. Rubber shall be in accordance with MIL-R-900 and shall be suitable for vulcanizing to steel strip (see 4.6). The hardness shall be 45 ± 5 durometer units (see 4.6.2).

3.2.2 Metal strip. The imbedded metal strip shall be a 3/64 by 1 inch hot rolled commercial quality steel strip, plated as necessary to secure adequate adhesion to rubber.

3.2.3 Adhesion (types A-1 and A-2). The adhesion of the rubber to the steel shall be such that the steel will bend or the rubber will tear before the rubber can be separated from the steel. It shall not be possible to strip the rubber cleanly from the steel without the use of solvent when tested as specified in 4.6.3.

3.2.4 Sealing pressure. The sealing pressure of the rubber shall be 95 pounds per square inch ($lb/in^2$) when tested in accordance with 4.6.4.

3.2.5 Recovered materials. Unless otherwise specified herein, all material incorporated in the gaskets covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recoverable materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

3.3 Design and construction.

3.3.1 Type A-1 straight section. Straight sections shall conform to figure 1.

3.3.1.1 At 5-inch intervals along the center of the strip and in the center of the 1/2-inch diameter recesses in the gasket, there shall be 1/4-inch diameter holes, countersunk for American National Standard No. 10 countersunk head machine screws as shown on figure 1.

3.3.1.2 Tolerances. The spacing of the holes in the metal strip shall be such that the distance between any two centers in a strip shall not vary from a multiple of 5 inches by more than plus or minus 1/64 inch. A tolerance of plus or minus 1/4 inch in length will be permitted. Tolerances in width and thickness will be permitted as follows:

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Width: plus 1/16 inch, minus 0.0 inch.

Thickness (total): plus 1/16 inch, minus 0.0 inch.

Thickness from ungrooved surface of rubber to metal strip: plus or minus 1/2 inch.

3.3.2 Type A-2, corner section. Corner sections shall conform to figure 2.

3.3.2.1 Tolerances. A tolerance of plus or minus 1/64 inch shall be permitted in the 6-5/16-inch dimensions, measured to the centers of the end holes in the metal strip. A tolerance of plus or minus 1/16 inch in the throat radius will be permitted. Tolerances in width and thickness shall be as follows:

Width: plus 1/16 inch, minus 0.0 inch.

Thickness (total): plus 1/16 inch, minus 0.0 inch.

Thickness from ungrooved surface of rubber to metal strip: plus or minus 1/32 inch.

3.3.3 Type B, two-line rubber. Type B shall be molded to the size shown on figure 3.

3.3.3.1 Tolerances. Tolerances shall be as shown on figure 3.

3.4 Workmanship. Rubber and metal shall be free from defects that might affect their serviceability; adhesion of rubber to metal shall be uniform throughout contacting surfaces; metal strips shall be centered; the ungrooved surface of gaskets shall be free from pits, indentations, projections, or bubbles; pits or indentations on the corrugated surface shall be not less than 5 inches apart in the lengthwise direction on types A-1 and A-2. Type B shall be free from imperfections.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).
(c) Inspection of packaging (see 4.7).



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4.3 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.3).

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, utilizing test specimens prepared from the material stock proposed to be used for construction of the gaskets. Three tests shall be performed:

(a) Hardness test (see 4.6.2).
(b) Adhesion test (see 4.6.3).
(c) Sealing pressure test (see 4.6.4).

One test specimen shall be provided for each test in accordance with the test methods specified herein.

4.4.1 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document included in the contract (see 6.2.3).

4.5 Quality conformance inspection.

4.5.1 Inspection lot. An inspection lot shall consist of all gaskets of the same type produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Sampling for visual and dimensional examination. From each lot, a random sample of gaskets shall be selected for visual and dimensional examination (see 4.6.1), in accordance with MIL-STD-105 for general inspection level II. The acceptance quality level (AQL) for major defects shall be 1.5 percent defective and the acceptance quality level for minor defects shall be 4.0 percent defective. The definitions of major and minor defects can be found in MIL-STD-105. MIL-STD-289 shall be used to classify defects in rubber gaskets.

4.5.3 Disposition of sample units. Sample gaskets which have been subjected to visual and dimensional examination (see 4.6.1) and determined to have one or more major defects shall not be offered for delivery to the Government.

4.6 Examinations and tests.

4.6.1 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.5.2 shall be visually and dimensionally examined to verify conformance with this specification. MIL-STD-289 shall be used to classify defects.

4.6.2 Hardness. The hardness shall be determined by test method ASTM D 2240 (see 3.2.1).

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4.6.3 Adhesion. Adhesion shall be tested in accordance with method B of ASTM D 429. The adhesion shall meet the requirements of 3.2.3.

4.6.4 Sealing pressure. Sealing pressure shall be determined by test method ASTM D 1081. The sealing pressure shall meet the requirements of 3.2.4.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Talc or talcum. Talc or talcum when used in the packaging process, for example, dusting of item(s), shall be asbestos free. A certificate of compliance shall be required that the dusting material is asbestos free (see 6.2.3).

5.1.2 Level A. Gaskets shall be individually unit protected in accordance with method III of MIL-P-116 as follows for the type gasket specified.

5.1.2.1 Type A-1. Straight sections shall be individually wrapped and provided with a stiffening media to prevent binding and distortion during handling, shipment and storage. The wrap shall be secured by taping or tying.

5.1.2.2 Type A-2. Corner sections shall be individually wrapped and provided with a stiffening media to prevent binding and distortion and further packaged in folding or set-up paperboard boxes. The number in each package shall be as specified (see 6.2.1). Folding boxes shall be closed by means of adhesive applied over not less than 95 percent of the area of contact between the flaps. Set-up and telescoping style boxes shall be closed with not less than 1-inch wide Kraft gummed tape. Box closures shall be strong enough to permit handling of box and contents without failure.

5.1.2.3 Type B. Two-line rubber material shall be furnished in rolls individually wrapped. The wrap shall be secured by taping or tying.

5.1.3 Level C. Packaging of gaskets shall afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractors normal retail or wholesale packaging methods may be utilized where such meets the requirements of this level.



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5.2 Packing. Packing shall be level, A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Gaskets, preserved-packaged as specified (see 6.2.1), shall be packed in containers conforming to any one of the following specifications at the contractor's option.

| Specification | Container | Type or Class |
|---|---|---|
| PPP-B-636 | Fiberboard | Weather-resistant |
| PPP-B-640 | Fiberboard, corrugated, triple-wall | Class 2 |

Containers shall be closed, waterproofed, and reinforced with tape or nonmetallic banding as specified in the appendix to the applicable specification, with method V closure applicable to PPP-B-636 containers. Containers shall not exceed the weight limitations of the applicable container specification.

5.2.2 Level B. Gaskets, preserved-packaged as specified (see 6.2.1), shall be packed in containers specified in 5.2.1, except that the containers shall be of domestic class of PPP-B-636 or class 2 of PPP-B-640. Containers shall be closed in accordance with the applicable container specification or appendix thereto, with method I closure applicable to PPP-B-636 containers, using pressure-sensitive tape.

5.2.3 Level C. Gaskets, packaged as specified, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing and method of shipment shall comply with Uniform Freight Classification Ratings, Rules and Regulations, or National Motor Freight Classification or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1) interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Types A-1 and A-2 gaskets are intended for use where tightness is desired between two flat surfaces, and are designed to be fastened by American National Standard No. 10 screws and cemented to hatch covers or ballistic doors by a nonhardening acetone cement. Type B gaskets are fitted to retainer strips and secured with adhesive, 1/8 inch by 5/8-inch flat bar and No. 10 screws.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and number or length (see 1.2).
(c) When first article is required (see 3.1).
(d) Level of preservation-packaging and packing required (see 5.1 and 5.2).
(e) Number of type A-2 gaskets per package (see 5.1.2.2).
(f) Special marking required (see 5.3).

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6.2.2 Types A-1 and A-2 gaskets should be purchased by the piece. Type B gaskets should be purchased in rolls of convenient lengths. Lengths of 100 feet are suggested. Since the strip can be bent to conform to corner radii, no provisions are made to furnish molded corner pieces for this type of gasket.

6.2.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664), and delivered in accordance with the approved CDRL incorporated in the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph no. | Data requirement title | Applicable DID no. | Option |
|---|---|---|---|
| 4.3 | Inspection system program plan | DI-R-4803 | ---- |
| 4.4.1 | First article inspection report | UDI-T-23790 | ---- |
| 5.1.1 | Certificate of compliance | DI-E-2121 | ---- |

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3.1 The data requirements of 6.2.3 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3.2 Test report data. The contract will specify that first article test reports shall be prepared by the manufacturer and sent to the contracting activity.

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material is acquired by the contractor for incorporation into the item and loses its separate identity when the item is shipped.



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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

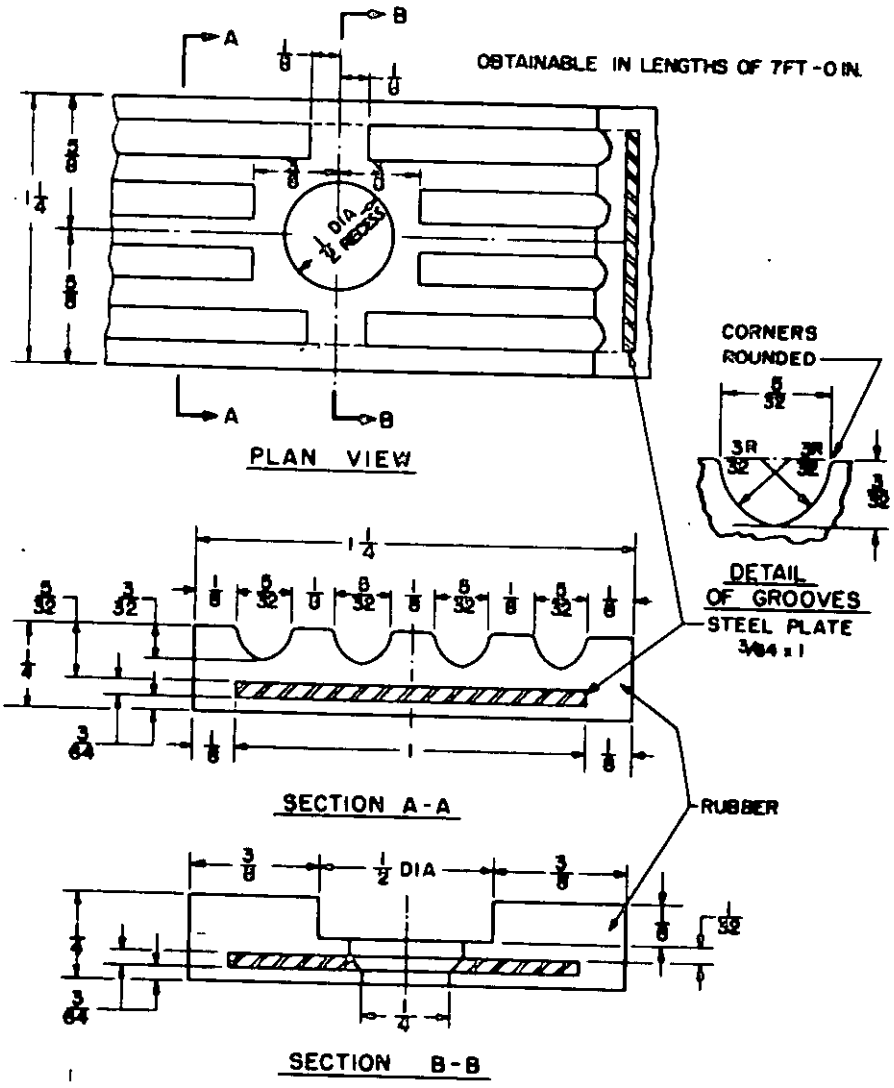
Custodians:
Army - AR
Navy - SH

Preparing activity:
Navy - SH
(Project 5330-0544)

Review activity:
DLA-IS



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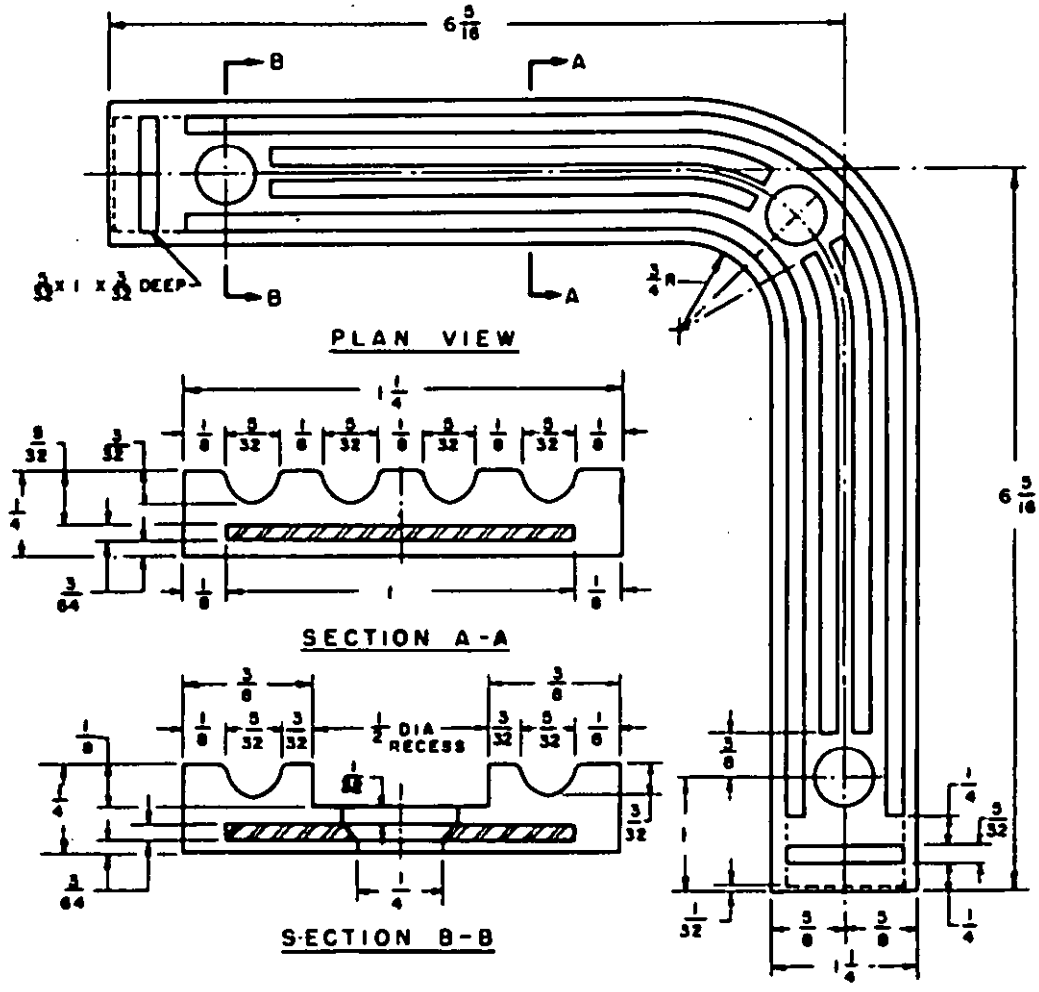

SH 1096

Notes: Except as noted, all dimensions are in inches.

FIGURE 1. Details of types A-1 and A-2.



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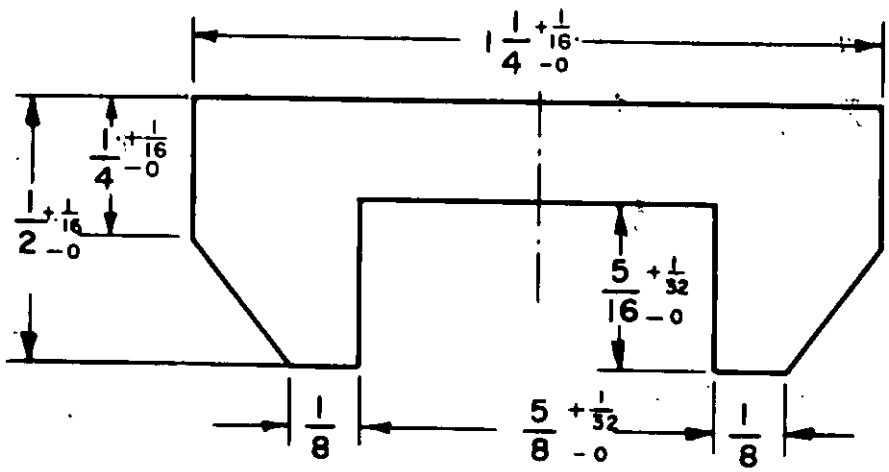



SH 1097

Note: For additional details of rubber gasket, see figure 1.

FIGURE 2. Detail of type A-2.

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Note: All dimensions are in inches.

FIGURE 3. Type B section.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-G-20078C | GASKETS FOR BALLISTIC DOORS AND HATCHES |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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